第二章作业

题量: 12　满分: 100

智能分析

作答时间: 10-12 10:22 至 10-27 13:00

一. 简答题（100分）

1 2 3

6 7 8

11 12

一. 简答题（共12题，100分）

1. (简答题)

2-1 How much bandwidth is there in 0.1 microns of spectrum at a wavelength of 1 micron?

我的答案:

The wavelength range is 0.95 to 1.05 micrometers. Using f = c/lambda, the bandwidth is Δf = 3×10^8/(0.95×10^-6) − 3×10^8/(1.05×10^-6) ≈ 3.01×10^13 Hz. Thus the bandwidth is approximately 3.0×10^13 Hz, or 30,000 GHz.

2. (简答题)

2-2 If a binary signal is sent over a 3-kHz channel whose signal-to-noise ratio is 20 dB, what is the maximum achievable data rate?

我的答案:

A 20 dB signal-to-noise ratio means S/N = 10^(20/10) = 100. The Shannon limit is 3000 log2(101) ≈ 19,975 bps. The Nyquist limit for a binary signal is 2B log2(2) = 2×3000 = 6000 bps. The lower limit applies, so the maximum achievable data rate is 6 kbps.

3. (简答题)

2-3 Ten signals, each requiring 4000 Hz, are multiplexed onto a single channel using FDM. What is the minimum bandwidth required for the multiplexed channel? Assume that the guard bands are 400 Hz wide.

我的答案:

Ten signals need nine guard bands. The minimum bandwidth is 10×4000 + 9×400 = 43,600 Hz = 43.6 kHz.

4. (简答题)

2-4 Suppose that A, B, and C are simultaneously transmitting 0 bits, using a CDMA system with the chip sequences of Fig. 2-22(a). What is the resulting chip sequence?

我的答案:

Sending a 0 means transmitting the negative of the station's chip sequence. Thus the combined sequence is −A − B − C = (+3, +1, +1, −1, −3, −1, −1, +1).

5. (简答题)

2-5 ACDMA receiver gets the following chips: (-1 +1 -3 +1 -1 -3 +1 +1). Assuming the chip sequences defined in Fig. 2-22(a), which stations transmitted, and which bits did each one send?

我的答案:

Let S = (−1, +1, −3, +1, −1, −3, +1, +1). The normalized inner products are S·A/8 = +1, S·B/8 = −1, S·C/8 = 0, and S·D/8 = +1. Therefore A and D sent 1, B sent 0, and C was silent.

6. (简答题)
2-6-1 Compare the delay in sending an x-bit message over a k-hop path in a circuit-switched network and in a (lightly loaded) packet-switched network. The circuit setup time is s sec, the propagation delay is d sec per hop, the packet size is p bits, and the data rate is b bps. Under what conditions does the packet network have a lower delay? Also, explain the conditions under which a packet-switched network is preferable to a circuit-switched network.

我的答案:

For circuit switching, T_circuit = s + x/b + kd. For store-and-forward packet switching, T_packet = x/b + (k−1)p/b + kd, assuming x ≥ p and negligible header, queueing and processing delays. Packet switching is faster when s > (k−1)p/b. It is preferable for bursty traffic, where links can be shared, or when alternate routes can provide fault tolerance after a switch or link failure.

7. (简答题)
2-6-2 Under what conditions does the packet network have a lower delay?

我的答案:

From T_circuit = s + x/b + kd and T_packet = x/b + (k−1)p/b + kd, the packet network has a lower delay when s > (k−1)p/b. This assumes negligible packet headers, queueing and processing delays.

8. (简答题)
2-6-3 Also, explain the conditions under which a packet-switched network is preferable to a circuit-switched network.

我的答案:

Packet switching is preferable for bursty traffic because users can share link capacity instead of reserving a circuit while idle. It is also preferable when circuit setup would add substantial delay, or when packets can be rerouted around a failed switch or link using an available alternative route.

9. (简答题)
2-7 Suppose that x bits of user data are to be transmitted over a k-hop path in a packet-switched network as a series of packets, each containing p data bits and h header bits, with x≫ p+h. The bit rate of the lines is b bps and the propagation delay is negligible. What value of p minimizes the total delay?

我的答案:

Approximately x/p packets are needed. The total delay is T(p) = [(p+h)/b] [x/p + k−1]. Differentiating gives dT/dp = (1/b)[−xh/p^2 + (k−1)]. Setting this to zero gives p^2 = xh/(k−1), so the minimizing data size is p = sqrt[xh/(k−1)] bits, for k > 1. This uses a continuous approximation.

10. (简答题)
2-8 A cable company decides to provide Internet access over cable in a neighborhood consisting of 5000 houses. The company uses a coaxial cable and spectrum allocation al lowing 100 Mbps downstream bandwidth per cable.

一. 简答题（100分）

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我的答案:

A 100-Mbps cable can guarantee 2 Mbps to at most 100/2 = 50 houses. For 5000 houses the company therefore needs 5000/50 = 100 separate coaxial segments, each connected to a fiber node. The upstream network must also have enough capacity to sustain the guarantees.

11. (简答题)

2-9 How long will it take to transmit a 1-GB file from one VSAT to another using a hub as shown in Fig. 2-50? Assume that the uplink is 1 Mbps, the downlink is 7 Mbps, and circuit switching is used with 1.2 sec circuit setup time.

我的答案:

The 1-Mbps uplink is the bottleneck. The propagation delay over four satellite legs is approximately 4×35,800/300,000 = 0.48 s. Using the slide's convention, the transmission time is 1024×8 = 8192 s. Including circuit setup, T = 1.2 + 8192 + 0.48 = 8193.68 s, about 2 h 16 min 34 s. If GB means 10^9 bytes with decimal Mbps, the result is approximately 8001.68 s; if the file is 1 GiB, it is approximately 8591.61 s.

12. (简答题)

2-10 Calculate the transmit time in the previous problem if packet switching is used instead Assume that the packet size is 64 KB, the switching delay in the satellite and hub is 10 microseconds, and the packet header size is 32 bytes.

我的答案:

Assume the file is 1 GiB, each packet carries 64 KiB of data plus a 32-byte header, link rates are decimal, and switching is store-and-forward. There are N = 2^30/(64×2^10) = 16,384 packets, each L = (65,536+32)×8 = 524,544 bits. The four link rates are 1, 7, 1 and 7 Mbps. Including the transmission pipeline, four propagation legs and three 10-microsecond switching delays, T = (N+1)L/(10^6) + 2L/(7×10^6) + 4×35,800/300,000 + 3×10^-5 ≈ 8595.28 s. If GB and KB are decimal and 64 KB is the data payload, the result is approximately 8005.14 s. The slide's claim of only 16 packets is incorrect.

一. 简答题（100分）

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